A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2,3,4,!,2,5,I,2,3,4,5(No of Frames=3).

CODE:

```
GNU nano 7.2                                lru.c *
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12, frames = 3;
    int frame[3], time[3];
    int i, j, k, pos, faults = 0, hits = 0, min;

    // Initialize frames
    for(i = 0; i < frames; i++) {
        frame[i] = -1;
        time[i] = 0;
    }

    for(i = 0; i < n; i++) {
        int found = 0;

        // Check hit
        for(j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                hits++;
                time[j] = i;
                found = 1;
                break;
            }
        }

        // If fault
        if(!found) {
            min = time[0];
            pos = 0;

            for(j = 1; j < frames; j++) {
                if(time[j] < min) {
                    min = time[j];
                    pos = j;
                }
            }

            frame[pos] = pages[i];
            time[pos] = i;
            faults++;
        }
    }

    printf("LRU Page Faults = %d\n", faults);
    printf("LRU Page Hits = %d\n", hits);

    printf("Hit Ratio = %.2f%%\n", (float)hits/n * 100);
    printf("Fault Ratio = %.2f%%\n", (float)faults/n * 100);

    return 0;
}

^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-] To Bracket  M-Q Previous  ^B Back      ^◂ Prev Word
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy      ^Q Where Was    M-W Next      ^F Forward   ^▸ Next Word
```

OUTPUT:

```
laxmi@laxmi: ~
laxmi@laxmi:~$ LRU.c
LRU.c: command not found
laxmi@laxmi:~$ nano lru.c
laxmi@laxmi:~$ gcc lru.c -o lru
laxmi@laxmi:~$ ./lru.c
-bash: ./lru.c: Permission denied
laxmi@laxmi:~$ ./lru
LRU Page Faults = 10
LRU Page Hits = 2
Hit Ratio = 16.67%
Fault Ratio = 83.33%
laxmi@laxmi:~$
```